GLACIS FIELD GUIDE

The offline penetration-testing worksheet & practical-exam companion

Version 0.2.0 “Pulse” · eJPTv2 / eCPPT friendly · works the same in labs, CTFs and client work

GLACIS is a keyboard-driven terminal worksheet that sits next to your tools and remembers everything for you: targets, ports, services, credentials, flags, proofs, dead ends and the commands you ran. It **never attacks anything itself** — it shows you the right command, copies it to your clipboard, and you run it in your own terminal. Everything lives in a local SQLite file. No network, no cloud, no AI in the loop.

The whole method in one line

SCAN › **RECORD** › **COPY** › **RUN** › **MARK** › **EXPORT**

Why it wins in practical exams	Exam-integrity compliance
• One screen answers: which host, what’s open, what’s next, what did I prove? • Ready-made commands per service — press `Enter` to copy, zero typos. • Credential matrix stops password-reuse amnesia across machines. • Pulse dashboard shows exactly which machine is being neglected.	• **100% offline** — local SQLite, zero network calls, zero telemetry. • **Zero autonomous action** — GLACIS copies commands; you execute them. • **No AI / cloud assistance** — nothing that violates exam rules. • Equivalent to permitted personal notes (Obsidian / CherryTree), just faster.

What’s in this guide

This guide was refreshed for v0.2.0 with documentation assistance from **GPT-6 Astra (medium)**; every command shown was verified against the v0.2.0 source.

1 Start in 10 minutes

Do this once per lab. Every step is optional except 1–3 — GLACIS never forces ceremony.

#	Step	Do exactly this
1	**Install & launch**	`pip install -e . # inside the repo` `glacis # opens the TUI`
2	**Make it yours**	`T # theme picker, d = save default` `:theme midnight # or direct switch`

3	**Create the lab workspace**	`W # workspace manager`
		`:ws init lab01 # scaffolds scans/ enum/`
		`# screenshots/ notes/ loot/`

4	**Get targets in fast**	`I # import nmap XML/txt/gnmap`
		`:t 10.10.10.20 # or add one by hand`
		`:subnet 10.10.10.20 10.10.10.0/24`

5	**Load a methodology**	`m # template picker` `:m ejpt # master exam checklist`
6	**Work the loop** (§3)	`j/k, Enter, Space # copy cmd, run it,` `# mark result, repeat`

7	**Export & protect**	`:export exam # markdown dossier`
		`glacis export --format html -o report.html`
		`glacis backup --label checkpoint`

Prefer plain shell? Every TUI action has a CLI twin — see the pocket reference in §10.

■ Mind-set

GLACIS is the co-pilot, not the pilot. It remembers, suggests syntax and tracks progress; **you** scan, exploit and decide. Write down everything the moment you learn it — future-you at hour six is a stranger.

2 The five stations

GLACIS splits the work across five screens. Switch with the number keys — anything you record is shared by all of them.

Key	Station	What it gives you	Go here when…
`0`	**Pulse** — engagement dashboard	Stat cards, 14-day momentum sparkline, per-target scorecards (A–F), next-action queue, live timeline of everything recorded.	You finish a box, feel lost, or wonder what to touch next.
`1`	**Cockpit** — the workbench	Target tree, services & ports with triage states, methodology checklist, notes and the command console.	Actually working a machine — this is home.
`2`	**Playbooks**	Offline command encyclopedia: per-service attack recipes you can copy with Enter.	You know the service, not the syntax.
`3`	**Credentials**	2D matrix: accounts × services. Compiles spray commands; tracks valid / pwned / invalid.	You hold creds and need to know where they work.
`4`	**Loot & Flags**	User/root flags, foothold & privesc proofs, question evidence, disk-loot browser, rabbit-hole log.	You captured something — or you’re stuck.

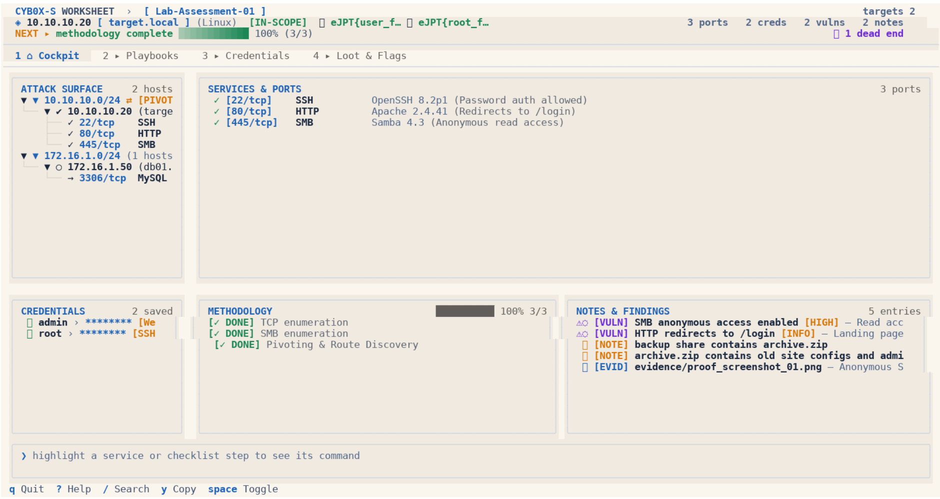
■ **New in v0.2.0 — Pulse, HTML reports and snapshots**

Press `0` anytime for the new **Pulse** dashboard: momentum sparkline, per-target scorecards and a next-action queue built from your own open items. Export a self-contained **HTML report** with `glacis export --format html`, and protect long sessions with `glacis backup`. All still 100% offline — see §9.

3 The core loop — six beats, every machine

Everything in GLACIS serves this loop. It takes seconds per beat and leaves a perfect trail: **what you tried, what worked, what to prove.**

Beat	Key / command	What happens
1 · RECORD	`:t 10.10.10.20 · :s 445/tcp smb`	Target and services land in the cockpit; checklists and playbooks key off them.
2 · COPY	**`j / k`** **`Enter`**	Highlight a service — the console shows a ready command with the IP substituted. Enter copies it.
3 · RUN	`(your terminal / tmux pane)`	Paste and execute. GLACIS never runs anything.
4 · MARK	**`Space`**	Cycle the service status: UNTESTED → CHECKED → DEAD-END → DEFERRED. The checklist % and Pulse coverage update.
5 · CAPTURE	`:c user:pass · :uflag … · :q 7 …`	Creds, flags and question proofs go straight into the ledger while you have them.
6 · PIVOT	**`]`**	Next target. Check station 0 if you’re unsure what “next” is.



Station 1 Cockpit — attack-surface tree (left), services with triage states (top right), methodology & notes (bottom), command console underneath.

■ PRO-TIP — the zero-mouse flow

Highlight a port with **`j / k`** → read the console → press **`.`** to swap tool recipes (whatweb → feroxbuster → gobuster → nikto → curl) → **`Enter`** copies → run it in tmux → **`Space`** to mark the result. Your hands never leave the keyboard.

4 Keyboard map — grouped by job

Single letters fire instantly unless a modal or the console input is focused. On day one you only need the eight keys in the blue box — the rest come naturally.

A · Survival — learn these first

Key	Does
`0 1 2 3 4`	Jump to Pulse / Cockpit / Playbooks / Credentials / Loot & Flags.
`j / k`	Move down / up in the focused list or tree (arrows work too).
`Enter`	Copy the highlighted command / value (matrix: compiled spray command).
`Space`	Cycle triage status; in station 3 cycle UNTESTED → VALID → PWN3D → INVALID; in station 4 preview loot.
`y`	Quick-copy the row’s core value (IP, IP:port, secret, note text).
`/ or Ctrl+F`	Global search across everything; Enter copies the top hit.
`?`	Full help & shortcut reference.
`q`	Quit (your data is already saved).

B · Capture — get findings in fast

Key	Does
`t / s / c / n`	Add target / service / credential / note via modal (or just type :t, :s, :c, :n in the console).
`f`	Record a finding with optional severity.
`K`	Add a custom checklist item (capital K — lowercase k moves).
`m`	Methodology template picker.
`I`	Import a scan: Nmap XML/text/gnmap, FFUF, feroxbuster, gobuster — staged for review before anything is committed.
`g`	Record captured user/root flags.
`a`	Link a question number to proof evidence (station 4).
`v`	Paste clipboard screenshot into screenshots/ and attach as evidence.

C · Move fast

Key	Does
`[ / ]`	Previous / next target — rotate machines in one keystroke.
`, / .`	Cycle the tool recipe for the highlighted service in the console.
`w · h / l`	Cycle panel focus; jump to left / right column.
`b · z`	Collapse sidebar · zoom focused panel full-screen.
`o · r · T`	Toggle scope · offline reference modal · theme picker (d saves; G toggles derived guidance, off by default).

> ■ **Day-one eight**
> `0-4` stations · `j / k` move · `Enter` copy · `Space` mark · `:` console · `y` copy value · `/` search · `?` help. Everything else can wait until you feel the rhythm — and `r` opens the offline syntax cheatsheet.

5 Console command reference

Press **:** and type. Tab completes; Esc returns to the workbench.

Capture

Command	Effect
`:t 10.10.10.20 [hostname] [os] [subnet] [pivot]`	Add target (also: `add target` … in plain words).
`:s 445/tcp smb [-- notes]`	Record an open service on the active target.
`:c admin:Secret123! [scope]`	Record a credential; scope tags the service.
`:c crack 3 Password123!`	Upgrade a captured hash (row 3) to plaintext in place.
`:n any free text`	Timestamped note attached to the active target.
`:f Suspicious ACL on /admin`	Finding with optional severity prompt.

Knowledge & navigation

Command	Effect
`:ref winrm · :ref smb`	Offline command reference for a service/tool.
`:m ejpt · :m web append`	Load methodology template (replace or append).
`:theme midnight`	Live theme switch (persist with T → d).

Host state, proofs & evidence

Command	Effect
`:uflag 7a9e… · :rflag f04b…`	Record user / root flag for the active target.
`:foothold smb null session · :privesc SUID …`	Record how you got in and how you went up.
`:q 7 /etc/shadow root hash`	Pin exam-question 7 to its proof string.
`:stuck wp-login brute = rabbit hole`	Log a rabbit hole (also :dead).
`:clue backup.zip in downloads share`	Log the breakthrough clue.
`:ev latest [desc] · :paste-ev [desc]`	Attach newest OS screenshot / paste clipboard image as evidence.
`:subnet 10.10.10.20 10.10.10.0/24`	Tag which subnet a target lives in.
`:pivot 10.10.10.20 172.16.1.0/24`	Mark a dual-homed pivot host + its hidden subnet.

Workspace & output

Command	Effect
`:ws lab02 · :ws init lab02`	Switch workspace / scaffold a new lab folder tree.
`:lhost auto · :lport 4444`	Attacker VPN IP (auto-detects tun0/wg0) & listener port — substituted into playbooks.
`:w rockyou · :w common · :w medium`	Copy standard wordlist paths to clipboard (see aliases below).
`:export wordlists`	Write deduplicated loot/users.txt + loot/passwords.txt.
`:export exam`	Markdown dossier: targets, question proofs, flags, creds, audit trail.
`:w <alias>`	Copy a wordlist path: `rockyou · common · medium · big / small · raft-d / raft-f · users / passwords · fasttrack`.

6 Workflow I — first contact with a machine

The repeatable 10-minute pattern for any new box. Left: what you run in your terminal. Right: what you tell GLACIS.

#	In your terminal	In GLACIS
1	`nmap -sn 10.10.10.0/24`	Spot the alive hosts.
2	—	`:t 10.10.10.20 hostname os · :subnet …`
3	`nmap -sS -p- --min-rate 1000 10.10.10.20`	— (or import whole XML next step)
4	`nmap -sV -sC -p 22,80,445 10.10.10.20 -oX out.xml`	`I → review staged hosts/services → commit`
5	—	`:m ejpt # methodology on this target`
6	run the copied recipe	**`j/k`** highlight service, **`.`** pick tool, **`Enter`** copy
7	`(paste & execute)`	**`Space`**
8	find creds / flag / foothold	`:c … · :uflag … · :foothold … · :q …`
9	stuck? move on	`:stuck … then ] # rotate to next target`

II Workflow II — the exam battle plan (5 phases)

Phase	Goal	GLACIS verbs
PHASE 1 · Scope & discovery	Find every machine and put it on the record.	`nmap -sn / arp-scan → :t + :subnet per host · :pivot for dual-homed boxes`
PHASE 2 · Enumerate everything	Full TCP sweep, then versions; import instead of typing.	`nmap -p- → -sV -sC -oX → I import · Space-mark each service as you clear it`
PHASE 3 · Low-hanging fruit	Anonymous & default access first, then web fuzzing.	`ftp/smb/snmp quick wins → :w common + feroxbuster → :f findings with severity`
PHASE 4 · Sprawl with creds	Every cred into the matrix; spray SMB/SSH/WinRM everywhere (§7).	`:c harvest → station 3 → Enter = spray cmd → Space = VALID/PWN3D`
PHASE 5 · Prove & submit	Flags, question proofs, clean dossier — before you forget.	`:uflag/:rflag · :q n proof · :export exam · glacis backup --label submitted`

■ THE 20-MINUTE RULE — the single highest-value habit

Stuck on one service for 20 minutes with no new lead? **Stop.** **`Space`** it to `DEFERRED`, type **`:stuck what-you-tried`**, press **`]`** and rotate. The Pulse queue and your DEAD-END trail will bring you back with fresh eyes. Most failed practicals are one 3-hour rabbit hole deep.

III Workflow III — the credential spray loop

Multi-machine exams are won on password reuse. Station 3 lays every credential against every service, so “did I try this password on this box?” always has a visible answer.

Step	Action
1 · HARVEST	Pull creds from web configs, DB dumps, SAM hive, shell history.
2 · RECORD	`c` `:c dbuser:S3cret! mysql`
3 · SPRAY	Station **3** → highlight a cred × service cell → `Enter` copies the compiled spray command.
4 · RUN & MARK	Run it; `Space` the cell: UNTESTED → VALID → PWN3D → INVALID (persisted).
5 · EXPLOIT THE WIN	PWN3D cell = your foothold on that box: back to station 1, capture flags, repeat.

Station 3 — the credential × service matrix. Badges persist in SQLite; restarting the TUI loses nothing.

> **■ Cracked a hash mid-session?**
> Don’t duplicate the row — upgrade it: `:c crack <row> <plaintext>`. Then `:export wordlists` writes deduplicated `loot/users.txt` + `loot/passwords.txt` for hydra / netexec runs.

IV Workflow IV — pivot into the internal network

Step	Action

1 · SPOT & ROUTE	`ip a`
	on a pwned box → second NIC = hidden subnet. Record:
	`:pivot 10.10.10.20 172.16.1.0/24`
	— the console switches to tunnel guidance.

2 · TUNNEL	`chisel server on pwned box → client on Kali → proxychains nmap -sT 172.16.1.0/24`
3 · RECORD THE REST	`:t 172.16.1.50 --notes via-pivot` — same core loop; the target tree shows both segments.
WHY IT PAYS	Exam questions ask about **hidden subnets and routes**, not just flags — and the dossier spells your routes out.

V Workflow V — prove it and submit

Evidence you didn’t write down is evidence you don’t have. Station 4 is where proofs live.

You captured…	Do this
A user/root flag	`:uflag <hash> · :rflag <hash>`
An exam answer	`:q <question#> <exact proof string> # or key a in station 4`
A screenshot	**v** pastes the clipboard image into screenshots/ and links it as evidence · `:ev latest desc`
Loot on disk	The loot browser indexes scans/, enum/, loot/, screenshots/ — Space previews, Enter copies the path.
A dead end	`:stuck … / :clue … # future-you says thanks`

Station 4 — flags, foothold & privesc proofs, question evidence, disk loot, rabbit-hole log.

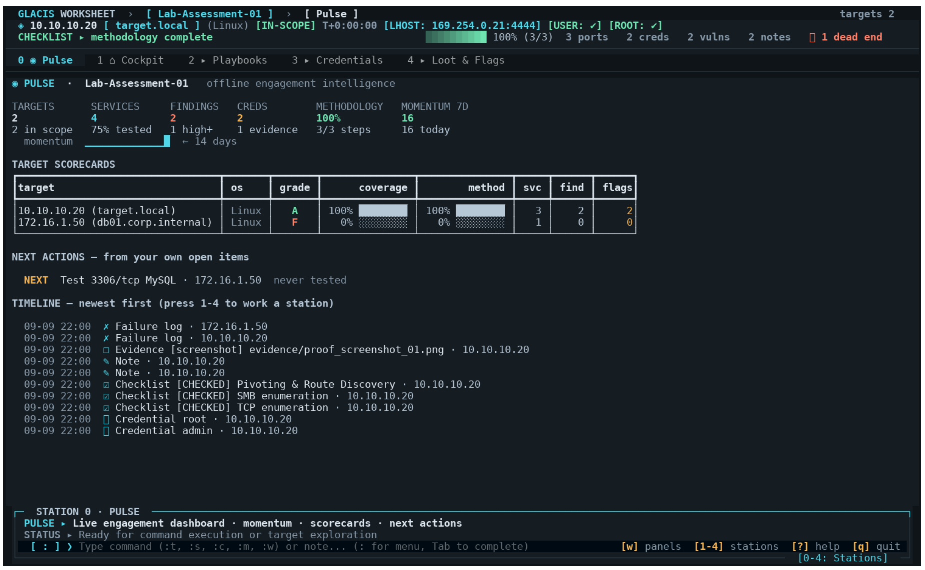
✓ The submission moment

Before touching the exam portal: **`:export exam`** → a self-contained Markdown dossier (target inventory, question proofs, flags, credential states, audit trail). Keep it open beside the portal and answer from it. Want something prettier to attach to a client report? **`glacis export --format html -o report.html`** renders a styled, self-contained HTML report (masked credentials, printable).

9 Pulse — see everything, choose smarter

Press `0`. Pulse is a read-only dashboard computed from what you already recorded: coverage (% services tested), methodology %, 14-day momentum sparkline, per-target scorecards with a transparent A–F grade, a **next-action queue** (your own untested services, missing proofs, open leads — ranked NOW / NEXT / LATER) and the newest timeline events.

Moment	What to look at
Just finished a box	Scorecards → the neglected machine (lowest coverage, no flags) is usually your next target.
Don’t know what’s next	The NOW/NEXT queue — it’s your own TODOs, ordered. Work top down.
Feels like slow progress	Momentum sparkline — flat two days = rotate sooner, log rabbit holes.
CLI equivalent	`glacis stats · glacis timeline -n 40`



Station 0 Pulse — stat cards, momentum, scorecards, next actions and timeline. Zero inference: it only counts what you recorded.

10 Templates, CLI pocket reference & exam day

Methodology templates (:m <name>)

Template	Focus	Covers
:m ejpt	Master flow	Scope → discovery → enumeration → foothold → pivoting → privesc — the exam spine.
:m discovery	Network	Local subnets, ARP/ICMP sweeps, TTL guesses, dual-homed discovery.
:m web	Web app	Headers, robots, dir fuzzing, SQLi, LFI, XSS, command injection, uploads.
:m smb	SMB	Null sessions, share perms, backups in shares, enum4linux, RID cycling.
:m pivoting	Routing	Dual-homed detection, autoroute, SOCKS5, SSH tunnels, chisel, proxychains.
:m linux · :m windows	PrivEsc	SUID/SGID, sudo -l, cron, capabilities / SeImpersonate, unquoted paths, AIE.
:m ftp · :m ssh · :m snmp	Services	Anonymous login, banner CVEs, key perms / communities, MIB walk.
:m databases · :m cracking	Data & hashes	Blank root logins, LOAD_FILE, xp_cmdshell / hash ID, John, Hashcat modes, hydra.

CLI pocket reference — when you’d rather stay in the shell

Command	Records / does
glacis t 10.10.10.20 -h dc01 --os Linux	Add a target.
glacis s 10.10.10.20 445/tcp SMB -v 'Samba 4.3'	Add a service.
glacis c admin:secret --source backup.zip	Add a credential.
glacis n 'checked share, nothing' · glacis f 'anon SMB' -s HIGH	Note / finding.
glacis import scan.xml	Staged scan import (same review as key I).
glacis search 'backup'	Search everything.
glacis stats · glacis timeline	Pulse from the shell.
glacis backup --label pre-enum · glacis backups	Snapshot / list snapshots.
glacis restore <snapshot.json>	Restores as a NEW workspace — never overwrites.
glacis export --format md|json|txt|html [-o file]	Export (html = styled report).

Exam-day checklist

First 15 minutes

1. Launch **glacis**, pick theme (T, d).
2. **:ws init exam** — clean workspace.
3. Import the scope/first scan (I).
4. **:lhost auto** · **:lport 4444**.
5. **:m ejpt** on the first target.
6. **glacis backup --label start**.

Last 30 minutes

1. Station 0: any target without flags? NOW queue empty?
2. Walk every DEFERRED service once more.
3. Every question has a :q proof? Check count.
4. **:export exam** → answer from the dossier.
5. **glacis export --format html -o exam.html**.
6. **glacis backup --label submitted**.

✓ Where your data lives

One SQLite file (default ~/.local/share/glacis/notebook.db, or .glacis/ beside an initialised workspace) plus plain folders for scans and loot. Copy them to a USB stick and you have a full backup — or let **glacis backup** rotate snapshots for you. No network, ever: safe for exam NDAs by design. Always defer to your certifying body’s current rules.